RapidBoxForest: Cross-Revision Endpoint-Envelope Reuse for Region-Graph Motion Planning

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***Abstract*—Safe-region and Graphs of Convex Sets (GCS) planners amortize queries over scene-dependent region graphs, whereas modified-environment roadmap methods retain or update sampled graph elements. RapidBoxForest (RBF) reuses a different object: robot-only endpoint enclosures keyed by canonical dyadic joint cells. Each address is both a Lifelong Envelope Cache Tree (LECT) key and a primitive of the scene graph rebuilt after a compatible obstacle revision. Replay therefore requires no geometric rematching and carries no obstacle label or adjacency across revisions. A budgeted builder refines these cells, coalesces only exact dyadic unions, and forms typed corridors from current-scene witnesses. On Shelf+IIWA, the warm profile reports a 0.171-s reusable build and a 0.077-s median within-batch task-completion diagnostic; all 40 paths pass an independent 0.01-rad sampled audit. Across three robots, amortizing the recorded build over ten rather than one query decreases the build-plus-task accounting coordinate by 69.6–79.7%. On the six-condition in-sample profile-selection catalog, every selected RBF profile returns audit-passing paths for all 100 instances in its condition. At those operating points, median query-matched path ratios are lower than the bidirectional rapidly-exploring random-tree planner (RRT-Connect) in six conditions and than probabilistic roadmaps (PRM) and Batch Informed Trees (BIT*) in five.**

***Index Terms*—Motion planning, robot manipulators, multi-query planning, collision checking, configuration-space decomposition, link envelopes.**

I. Introduction

Manipulators repeatedly solve related queries in semi-static workcells, where kinematics stay fixed but start–goal pairs and local clutter change. A reusable planner must amortize free-space construction, separate persistent kinematic evidence from scene-local validity, and track whether each returned segment has conservative current-scene support.

Existing methods occupy different reuse regimes. Probabilistic roadmaps, experience graphs, and safety certificates reuse connectivity, paths, or clearance neighborhoods, whereas tree and batch planners typically construct query-local search structures [1]–[3]. Certified polyhedral decompositions, safe-box planners, and Graphs of Convex Sets (GCS) methods instead make collision-free regions explicit [4]–[7]. This motivates the design question: how can explicit regions be reused across queries while obstacle-independent kinematic evidence—but no scene-validity claim—persists across compatible revisions? Table I compares the proposed RapidBox-Forest (RBF) and Lifelong Envelope Cache Tree (LECT) with representative planning families in terms of reusable state and validation responsibility.

Canonical dyadic identity aligns each LECT record with a primitive forest cell, eliminating geometric rematching during replay, batched validation, and refinement. Only obstacle-independent endpoint evidence persists; each revision rebuilds link envelopes, labels, and connectivity. Coalesced scene nodes remain dyadic-range aligned but never become cache keys.

a) Problem statement: Let $\mathcal{Q} \subset \mathbb{R}^n$ be the joint-limit box of a fixed n-degree-of-freedom manipulator, and let $\{\mathcal{O}^{(r)}\}$ denote workspace obstacle revisions. Let endpoint-record identity κ_E identify the robot endpoint model and canonical dyadic decomposition, and let descriptor δ collect the active-link geometry and validator settings. For the current δ, $\mathcal{C}_{\text{free}}(\mathcal{O}^{(r)}) \subseteq \mathcal{Q}$ denotes the corresponding modeled free space; its dependence on δ is suppressed below. Each query supplies $(q_s, q_g) \in \mathcal{Q}^2$, and RBF builds a finite graph of policy-accepted C-space boxes. Queries within one revision reuse this graph. Across revisions, only endpoint enclosures with an exact joint-region and κ_E match are replayed; link envelopes, labels, and connectivity are regenerated. The target is a sound, resource-bounded volumetric planning structure.

b) Reuse and certificate semantics: RBF keeps persistent endpoint geometry separate from revision-local graph state and route-witness status. Table II states the returned-object contract.

The paper makes three contributions:

1) *Cross-revision evidence interface.* An identity invariant binds each endpoint record to the forest's canonical dyadic cell, separating revision-persistent robot geometry from state that must be rebuilt.
2) *Cache-aligned construction.* A resource-bounded builder replays or materializes endpoint evidence, refines unresolved canonical cells, and coalesces integer-verified exact unions without remapping primitive evidence keys or enlarging the verified set.
3) *Witness-typed corridor contract.* A returned-path contract composes overlap-box and finite-cover oriented bounding box (OBB) witnesses and separates them from tolerance-gap, segment, and repair transitions that remain validation-required.

Sections III–V formalize this pipeline (Fig. 1); Q1–Q4 characterize warm-profile costs, selected planning trade-offs, and envelope mechanisms.

II. Related Work

Region-based planners make collision-free volume explicit. IRIS (Iterative Regional Inflation by Semidefinite programming) constructs convex regions, and configuration-space IRIS (C-IRIS) certifies C-space polytopes [4], [8]. Graphics processing unit (GPU) methods accelerate region growth [9],

TABLE I
REUSABLE STATE IN REPRESENTATIVE PLANNING PARADIGMS

Paradigm family	Reused state	Explicit region cells	Update or validation style	Multi-query implication
Roadmaps, experience graphs, and clearance caches	Vertices, edges, path fragments, or clearance neighborhoods	Method-dependent	Recheck, retrieve, or repair affected graph elements	Strong connectivity or experience reuse without a persistent scene decomposition.
Tree and anytime sampling	None across queries; query-local tree or batch sample graph	Not in the standard formulation	Primarily query-local; rebuilt or checkpointed	Fast discovery or quality improvement without persistent volumetric scene structure.
Region construction and region-graph planning	Convex cells or safe boxes and their graph	Yes, for the represented scene	Construct, certify, optimize through, or recertify regions	Expressive or optimization-ready region reuse once a region collection exists.
RBF	Within-scene typed box graph; cross-revision obstacle-independent endpoint enclosures	Yes	Relabel boxes; rebuild adjacency; replay only matched endpoint evidence	Shared dyadic evidence keys with current-scene corridor verification.

[10], and visibility-graph clique covers reduce set count [11]. Fast safe-box planning preprocesses a safe-region collection [5], while GCS optimizes trajectories over convex-region graphs [6], [12]. Implicit GCS search (IxG/IxG*) and GCS* restrict online work to query-relevant portions [13], [14]; multi-query GCS precomputes a cost-to-go surrogate for repeated queries on a fixed GCS [7]. These methods accelerate search once the convex sets and successor relation are defined. RBF addresses the construction-and-update problem: it builds a budgeted dyadic graph whose primitive cells also key robot-only endpoint bounds, then recomputes all scene-dependent labels and overlaps after a revision.

a) Closest methods: ReUse-based probabilistic roadmap (RU-PRM) reuses obstacle-local roadmaps by matching and transforming models across similar environments [15]; Ball Tree attaches volumes to a single-query tree [16]. Safety certificates cache local clearance [2]; dynamic roadmaps, decomposed checking, and Simple Polyhedral Intersection Techniques (SPITE) accelerate updates of a pre-existing sampled graph; SPITE associates configurations and edges with workspace swept-volume approximations and intersects them with moved obstacles [17]–[19]. Experience graphs retain validated path or graph structure [3], while soft subdivision targets conservative cell classification [20]. The key distinction is the state that survives a revision. C-IRIS/GCS operate on scene-defined regions; RU-PRM and dynamic-roadmap/SPITE methods retain or update sampled graph elements; clearance caches retain local validity neighborhoods. RBF carries none of those scene-validity or connectivity claims across revisions. It reuses robot-only endpoint enclosures on canonical volumetric cells and reconstructs the current labels, overlaps, and corridor witnesses.

b) Endpoint bounds: Proximity queries use the Gilbert–Johnson–Keerthi (GJK) algorithm [21] and Flexible Collision Library (FCL) [22]. Interval methods bound nonlinear robot kinematics [23], [24]. For articulated continuous collision detection (CCD), interval hierarchies, swept primitives, and Taylor models bound motion [25], [26]. RBF separates versioned endpoint evidence from current link envelopes and scene separation.

c) Graph reuse: PRM and LazyPRM amortize or defer sampled-edge validation [1], [27]. Effort Informed Roadmaps (EIRM*), decomposed checking, and Repetition Roadmaps reuse prior edges, collision tests, or paths [18], [28], [29]. Adaptive cell planners refine accepted-cell graphs [20], [30], [31]. In contrast, RBF uses canonical volumetric cells both to index reusable robot-only endpoint bounds and to form a freshly labeled overlap graph under an explicit budget.

d) Sampling and refinement: RRT-Connect is a bidirectional single-query planner [32]; BIT* is an informed, anytime, asymptotically optimal sampling-based planner [33]. The Open Motion Planning Library (OMPL) supplies the benchmarking infrastructure [34]. RBF instead supplies reusable volumetric context before repair, optimization, and final audit.

III. Link Interval Envelopes

The endpoint/link split enables cross-revision reuse. For a joint region $I \subseteq \mathcal{Q}$, $E_k(I)$ bounds endpoint k's robot-specific trajectory and is reusable under an identity match, whereas $B_\ell(I)$ is derived under the current descriptor and tested against the current scene:

$$I \mapsto \{E_k(I)\} \mapsto \{B_\ell(I)\}.$$

RBF uses axis-aligned boxes as global forest cells and OBB regions only as local finite-cover corridor witnesses. Either supports certificate status only after conservative current-scene validation.

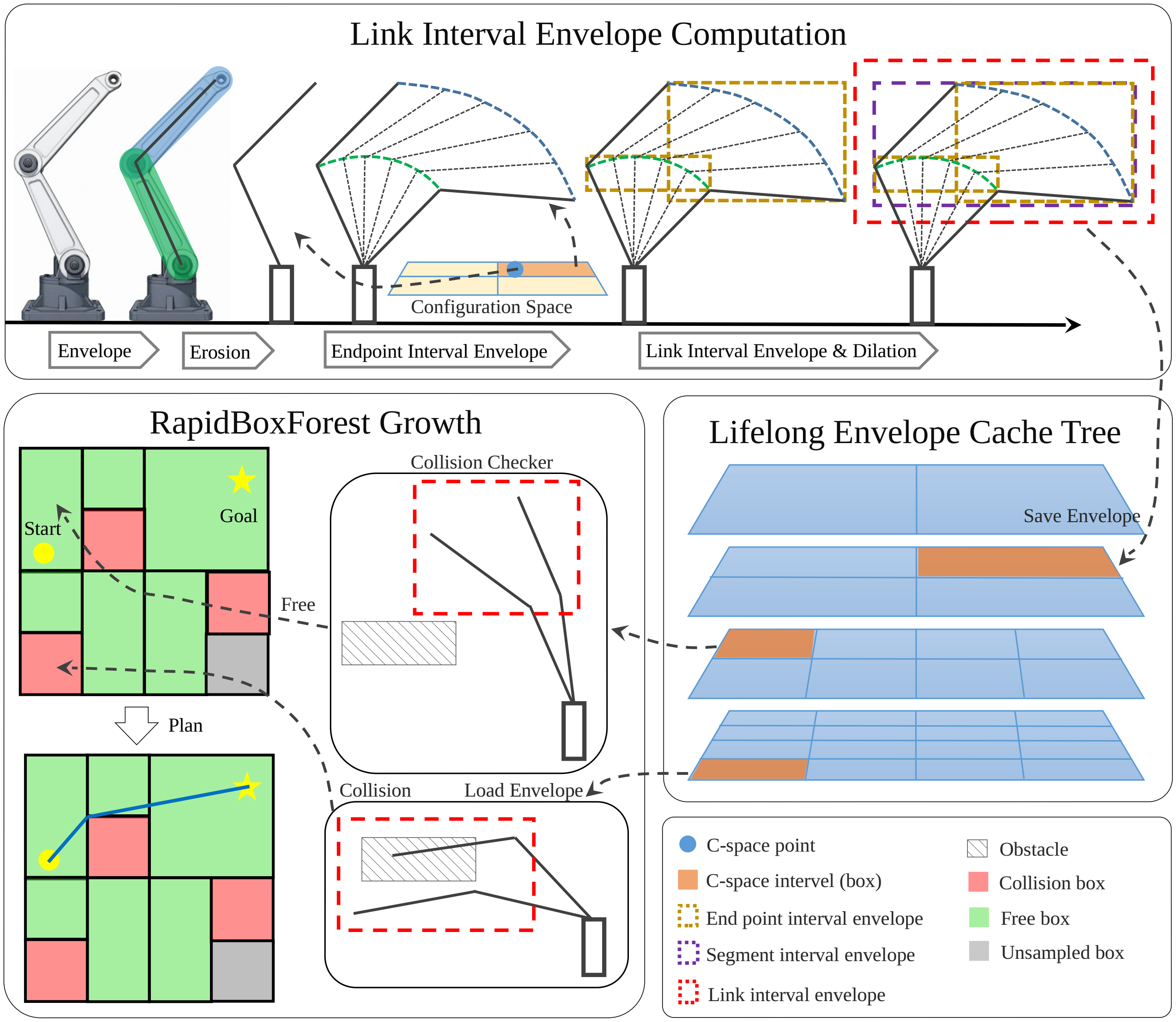


Fig. 1. RBF pipeline. Endpoint envelopes induce zero-radius centerline envelopes, which are expanded by the link radius (top). The lower-left panel shows sample-guided growth with accepted, rejected, and unsampled boxes; box statuses are defined in Table II. The lower-right panel shows LECT replay of endpoint evidence with recomputed obstacle labels and graph edges.

TABLE II
THEORETICAL RETURNED-OBJECT STATUS UNDER ASSUMPTION 1; EXPERIMENTAL ARTIFACTS FOLLOW THE VALIDATION-REQUIRED PROTOCOL IN SECTION VI

Object or state	Status	Condition
Finite box-covered path	Certificate-backed	The retained regions cover the path, and each returns the conservative-free status CertFree or inherits through verified, same-revision, descriptor-matched exact-union tiling.
Finite oriented bounding box (OBB) path	Certificate-backed	The retained OBB regions cover the path, and each complete region passes the affine-zonotope validator under the current revision and descriptor.
Tolerance-gap, segment, or repair edge	Validation-required	Requires conversion to a conservatively validated box or OBB corridor.
Path relying on filter-only evidence	Validation-required	Empirical evidence may guide search but does not certify the path.
LECT replay record	No path claim	The endpoint-evidence identity must match; scene labels and edges are recomputed.

Assumption 1 (Collision model for certificates)**.** *The certificate model covers active rounded links against a finite union $\mathcal{O}$ of closed workspace axis-aligned bounding-box (AABB) obstacles; boundary contact is collision, and $I \subseteq \mathcal{Q}$ enforces joint limits. A change to either the scene revision or descriptor δ invalidates every scene-dependent box and edge label. The status* CertFree *denotes this conservative free outcome; no other validator outcome carries a free-space claim. Self-collision, attached bodies, grippers, and task constraints require another conservative validator.*

a) Numerical contract: Soundness assumes outward-rounded endpoint and link envelopes and all margins in δ. Here $\mathbb{B}_p(r)$ is the p-norm ball of radius r; the AABB broadphase inflates by $\mathbb{B}_\infty(r_\ell)$, which contains the Euclidean link-radius ball. In the evaluated implementation, CertFree names the intended conservative branch; because its arithmetic was not independently verified and complete witnesses W were not retained, experimental paths are audited and reported as validation-required.

Definition 1 (Returned-path certificate status)**.** *Let ν_{scene} denote the current scene revision, let $\mathcal{O}(\nu_{\text{scene}})$ be its obstacle set, and fix descriptor δ and a continuous returned path $\gamma : [0,1] \to \mathcal{Q}$. A retained construction witness $W = \langle (t_{j-1}, t_j, R_j) \rangle_{j=1}^m$ satisfies $0 = t_0 < \cdots < t_m = 1$ and $\gamma([t_{j-1}, t_j]) \subseteq R_j$. Each retained R_j is either (i) an axis-aligned box for which the current validator returns* CertFree*, (ii) an axis-aligned box equal to a retained exact union of same-revision, descriptor-matched* CertFree *boxes with complete tiling provenance, or (iii) a complete C-space OBB region for which the affine-zonotope validator returns* CertFree *under that pair. The path is certificate-backed exactly when such a retained W exists; otherwise it is validation-required.*

A. Geometric Foundation

For a body-frame shape C_0, let $T(q)x = R(q)x + t(q)$ map point x to workspace, with rotation $R(q)$ and translation $t(q)$, and let $\mathcal{S}_I(C_0)$ denote its sweep over $q \in I$ and $\oplus$ the Minkowski sum. For a capsule $C = S \oplus \mathbb{B}_2(r)$ with center segment S, rigid motion gives

$$\mathcal{S}_I(C) = \mathcal{S}_I(S) \oplus \mathbb{B}_2(r).$$

Thus a rounded link can be bounded by first bounding the zero-radius center segment and then lifting by the link radius. Let $\Gamma_a(I)$ and $\Gamma_b(I)$ denote the exact workspace trajectories of a capsule link's proximal and distal endpoints over I. RBF stores conservative endpoint bounds in reusable records keyed by kinematics; the following proposition converts them into whole-link envelopes.

Proposition 1 (Endpoint envelopes induce link interval envelopes)**.** *Consider a rounded link with body-frame endpoints a, b, $L = [a, b] \oplus \mathbb{B}_2(r)$, and any joint region $R \subseteq \mathcal{Q}$. If endpoint envelopes $E_a(R)$ and $E_b(R)$ satisfy $\Gamma_a(R) \subseteq E_a(R)$ and $\Gamma_b(R) \subseteq E_b(R)$, then*

$$\mathcal{S}_R(L) \subseteq \operatorname{conv}\big(E_a(R) \cup E_b(R)\big) \oplus \mathbb{B}_2(r).$$

Consequently, any conservative outer representation of the right-hand side is a valid link envelope for the whole joint region R.

Proof. For each $q \in R$, the moved center segment is the convex hull of its endpoints. Hence the union of these segments lies in $\operatorname{conv}(\Gamma_a(R) \cup \Gamma_b(R)) \subseteq \operatorname{conv}(E_a(R) \cup E_b(R))$. Adding the fixed-radius ball then yields the capsule sweep and the stated containment. $\square$

Corollary 1 (Conservative configuration-space box validation)**.** *Let $I \subseteq \mathcal{Q}$ be an axis-aligned configuration-space box and let $\{B_\ell(I)\}$ be conservative link interval envelopes for all active rounded links. If $B_\ell(I) \cap \mathcal{O} = \emptyset$ for every link ℓ, then every configuration $q \in I$ is collision-free with respect to the collision model in Assumption 1.*

Proof. By conservatism, each $B_\ell(I)$ contains the complete workspace sweep of link ℓ over I. Its disjointness from $\mathcal{O}$ therefore excludes the modeled workspace-obstacle collision for every active link and every $q \in I$ under Assumption 1. $\square$

B. Endpoint Interval Envelopes

The cached primitive is an interval AABB (iAABB) enclosing the endpoint trajectory $\Gamma_k(I)$ and shared by all link representations. Affine-arithmetic interval forward kinematics (IFK-AA) uses one Denavit–Hartenberg interval pass. Hierarchical interval forward kinematics (HIFK) bisects I to depth d and hulls the child boxes, denoted HIFK-d. Critical-sample is filter-only; Q4 also reports Analytical and Monte Carlo empirical references. Fig. 1 summarizes the conversion.

C. C-space OBB Endpoint Envelopes

Axis-aligned boxes remain the global forest primitive; an OBB is local edge evidence for a bridge or repair. Let q_c be its center, let the diagonal matrix $S \succ 0$ scale the joints, let $U \in \mathbb{R}^{n \times m}$, $1 \le m \le n$, have orthonormal columns in scaled coordinates, and let $h \in \mathbb{R}^m_{\ge 0}$ contain the half-widths. With $A = S^{-1} U \operatorname{diag}(h)$ and $\Xi = \{\xi \in \mathbb{R}^m : -1 \le \xi_i \le 1\}$, the joint-space region is

$$R_{\text{obb}} = \{\, q_c + A\xi \mid \xi \in \Xi \,\}.$$

For R_{obb}, the correlated validator encloses each endpoint by the outward affine zonotope $E_k(R_{\text{obb}})$ specified below. Certificate eligibility requires $R_{\text{obb}} \subseteq \mathcal{Q}$, verified containment of the represented path window, and conservative validation of the complete region; the fitted orientation alone provides no safety status.

a) Finite path cover: A piecewise-linear candidate path γ has an OBB witness only if finitely many closed windows J_τ cover $[0,1]$, with $\gamma(J_\tau) \subseteq R_\tau \subseteq \mathcal{Q}$, and every complete R_τ passes the correlated endpoint/link test. An uncovered window remains validation-required; sampled fallbacks are search-only.

Corollary 2 (Finite-cover OBB-path soundness)**.** *Fix a scene revision ν_{scene} and descriptor δ, and write $\mathcal{O} = \mathcal{O}(\nu_{\text{scene}})$. Let $[0,1] = \bigcup_{\tau=1}^{M} J_\tau$ be a finite window cover of a continuous path γ, with $\gamma(J_\tau) \subseteq R_\tau \subseteq \mathcal{Q}$. Suppose that, for every τ, conservative affine-zonotope endpoint enclosures induce conservative envelopes for all active links over the complete region R_τ, and that those link envelopes pass current-scene separation under the same revision and descriptor. Then $\gamma([0,1]) \subseteq \mathcal{C}_{\text{free}}(\mathcal{O})$ under Assumption 1.*

Proof. For each τ, Proposition 1 and current-scene separation give $R_\tau \subseteq \mathcal{C}_{\text{free}}(\mathcal{O})$. For any $t \in [0,1]$, the window cover supplies an index τ with $t \in J_\tau$, and hence $\gamma(t) \in R_\tau \subseteq \mathcal{C}_{\text{free}}(\mathcal{O})$. The claim follows. □

b) Correlated endpoint envelope: For $q = q_c + A\xi$, outward affine propagation gives

$$E_k(R_{\text{obb}}) = c_k + \sum_{i=1}^{p} g_{k,i}\epsilon_i \oplus [-\rho_k, \rho_k].$$

Here p is the number of affine generators, $c_k, g_{k,i} \in \mathbb{R}^3$, $\epsilon_i \in [-1,1]$, and $\rho_k \in \mathbb{R}^3_{\geq 0}$; $[-\rho_k, \rho_k]$ is componentwise and ρ_k outwardly encloses the entire nonlinear remainder. A sampled remainder is filter-only. For workspace direction $d \in \mathbb{R}^3$, the support of E_k is

$$h_{E_k}(d) = d^\top c_k + \sum_{i=1}^{p} |d^\top g_{k,i}| + \sum_{\mu=1}^{3} |d_\mu| \rho_{k,\mu}.$$

The rounded-link support bound is therefore

$$\bar{h}_\ell(d) = \max\{h_{E_a}(d), h_{E_b}(d)\} + r_\ell \|d\|_2.$$

Coordinate directions provide the AABB broadphase; retained supports provide the SupportHull/GJK test. OBB records remain scene-local typed-edge evidence and do not become persistent LECT records or global forest nodes.

D. Link-Envelope Representations

Endpoint records can be exported to multiple link representations without changing the upstream kinematic source. For a serial-chain capsule link ℓ whose proximal and distal endpoint records are $E_{\ell-1}(I)$ and $E_\ell(I)$, $B_\ell(I) = B^\circ_\ell(I) \oplus \mathcal{R}(r_\ell)$, with $B^\circ_\ell(I)$ bounding the centerline sweep and $\mathcal{R}(r_\ell) \supseteq \mathbb{B}_2(r_\ell)$ denoting a conservative radius-lift set. AABB uses $\mathbb{B}_\infty(r_\ell)$, while SupportHull uses the Euclidean-ball support.

a) AABB envelope: AABB envelopes wrap the two endpoint envelopes in one box and pad by the link radius:

$$B_\ell(I) = \mathrm{bbox}\big(E_{\ell-1}(I) \cup E_\ell(I)\big) \oplus \mathbb{B}_\infty(r_\ell).$$

Here $\mathrm{bbox}(\cdot)$ denotes the workspace AABB hull.

b) SupportHull envelope: SupportHull stores support mappings for both endpoint enclosures and the link radius. The default implementation uses endpoint AABBs; correlated OBB corridors use the zonotope-plus-box records above. A bounded-work GJK narrow phase uses outward obstacle inflation. If the iteration limit is reached without a decision, the method returns $\mathrm{Unknown}$ rather than a free label.

Fig. 2 illustrates the resulting build, query, and update cycle on a two-link schematic.

IV. Lifelong Envelope Cache Tree

LECT supplies the cross-revision half of the reuse contract by keying obstacle-independent endpoint evidence to dyadic joint boxes shared with forest construction. RBF derives link envelopes under the current descriptor and refreshes every scene-dependent label; graph connectivity and query outcomes are never reused across scene revisions.

A. Cache Key and Split Policy

LECT is a dynamically grown binary k-d tree over $\mathcal{Q}$; each node stores a joint interval, split metadata, and geometry records.

a) Cache identity: Replay requires exact matches of endpoint-record identity κ_E and dyadic address. The identity binds robot kinematics, endpoint representation, root canonicalization, split policy, schema, and builder version, whereas δ binds link geometry and validation settings; only endpoint records persist. LECT and the forest follow one deterministic midpoint split schedule, aligning records with primitive cells without geometric rematching.

Formally, let κ_E pass the endpoint-record gate above, fixing a common root box, canonicalization, and split schedule s_{κ_E}. For a root-relative child address $\alpha = (c_1, \ldots, c_d)$, $c_i \in \{0,1\}$, the shared decoder maintains

$$I^{\text{LECT}}_{\kappa_E}(\alpha) = I^{\text{forest}}_{\kappa_E}(\alpha) = I_{\kappa_E}(\alpha).$$

For endpoint k, each stored record $(\kappa_E, \alpha, k, E_k)$ satisfies the record invariant

$$\Gamma_k(I_{\kappa_E}(\alpha)) \subseteq E_k.$$

Scene-local coalesced nodes store their verified tiling status and matched revision–descriptor label but never become persistent cache keys. This identity invariant, rather than a geometric lookup, interfaces cache replay and forest refinement.

b) Sparse interval staging: Planning may stage an exact interval without materializing every ancestor. If its endpoint record is absent, that interval is materialized under κ_E before validation under the current descriptor.

B. Runtime Use and Validation Semantics

For a joint interval, LECT replays or materializes endpoint evidence, derives link envelopes under δ, and applies the current-scene validator. When child endpoint records are hulled into a parent endpoint envelope, the resulting parent box must still pass the same validator. Missing or inconclusive evidence yields $\mathrm{Unknown}$. Cache hits may change enclosure tightness and hence the acceptance outcome, but not the meaning or validation obligations of any returned status.

The cache-identity gate above controls replay eligibility; safety then follows from decoded containment and fresh scene validation.

Proposition 2 (Sound replay of matched endpoint enclosures)**.** *Let $I = I_{\kappa_E}(\alpha) \subseteq \mathcal{Q}$, and fix the current scene revision ν_{scene} and descriptor δ. Suppose each required endpoint record matches the current κ_E and satisfies the record invariant above at address α. If the link envelopes derived under δ are conservative and all pass separation from $\mathcal{O}(\nu_{\mathrm{scene}})$, then $I \subseteq \mathcal{C}_{\mathrm{free}}(\mathcal{O}(\nu_{\mathrm{scene}}))$ under Assumption 1.*

Proof. The matched record invariant supplies every endpoint inclusion over I. Proposition 1 therefore places each rounded-link sweep inside its derived conservative envelope. Current-scene separation and Corollary 1 give the stated inclusion. □

C. Storage and Revision Semantics

Serialization retains compatible endpoint geometry and split metadata only; link envelopes, labels, adjacency, and query outcomes are reconstructed per revision.

V. RapidBoxForest Construction

Fig. 2 summarizes the planning layer. Shared dyadic cell identities let replay, batch validation, and refinement address the same joint boxes. RBF turns envelope validation and LECT replay into a query-reusable, scene-specific structure comprising policy-accepted C-space boxes and typed corridor adjacencies. Later queries use box membership and graph search; every accepted box is checked against current obstacles or inherits status through a same-revision exact-union witness.

A box is an axis-aligned joint interval $B = [l^{(1)}, u^{(1)}] \times \cdots \times [l^{(n)}, u^{(n)}] \subseteq \mathcal{Q}$. A *policy-accepted box* has passed the configured scene-validation policy. Under the stated geometric and numerical assumptions, conservative link-envelope disjointness makes it eligible for $\mathrm{CertFree}$; filter-only acceptance yields a validation-required region. Rejected or unresolved cells may be retained for refinement but are excluded from the forest.

A. Budgeted Construction Contract

Construction returns a resource-bounded reusable region graph. Given the joint-limit box $\mathcal{Q}$ and obstacle set $\mathcal{O}$, the scene policy $V_{\mathcal{O}}(B;\delta)$ evaluates active-link envelopes for B under descriptor δ and has range

$$V_{\mathcal{O}}(B;\delta) \in \{\mathrm{CertFree}, \mathrm{ProvisionalFree}, \\ \mathrm{Occupied}, \mathrm{Unknown}\}.$$

By definition of the conservative branch,

$$V_{\mathcal{O}}(B;\delta) = \mathrm{CertFree} \implies B \subseteq \mathcal{C}_{\mathrm{free}}(\mathcal{O}).$$

$\mathrm{ProvisionalFree}$ is filter-only, $\mathrm{Occupied}$ is a policy rejection, and $\mathrm{Unknown}$ is inconclusive. The accepted region set of N boxes is

$$\mathcal{F} = \big\{(B_i, \sigma_i) : i = 1, \ldots, N\big\}, \\ B_i \subseteq \mathcal{Q}, \qquad \sigma_i \in \{\mathrm{CertFree}, \mathrm{ProvisionalFree}\}.$$

The evaluated profile and Algorithm 1 retain only $\mathrm{CertFree}$ boxes; $\mathrm{ProvisionalFree}$ belongs to the general search-policy range but cannot contribute to W without conservative revalidation. For a same-revision, descriptor-matched family $\mathcal{G}$, let $H = \mathrm{bbox}(\bigcup_{B \in \mathcal{G}} B)$; the hull inherits this label exactly when

$$\mathrm{ExactUnion}(H, \mathcal{G}) \iff H = \bigcup_{B \in \mathcal{G}} B$$

holds. Let ε_c denote the connection tolerance, with $\varepsilon_c \geq 0$; it does not enter this predicate. Exact union is tested by dyadic integer-range coverage, not floating-point volume equality, and is enabled in Q1 but disabled in Q3. Let $\mathcal{S} \subseteq \mathcal{Q}$ be a finite anchor set fixed before queries; it affects admission priority, not $V_{\mathcal{O}}$ or certificate status. The reusable scene state $(\mathcal{F}, G)$ records revision/descriptor and exact-union provenance.

For two boxes

$$B_i = \prod_{d=1}^{n} [l_i^{(d)}, u_i^{(d)}], \qquad B_j = \prod_{d=1}^{n} [l_j^{(d)}, u_j^{(d)}],$$

RBF treats the boxes as candidate-adjacent when the closed intervals in every coordinate overlap or have a gap no larger than ε_c:

$$\max\{l_i^{(d)}, l_j^{(d)}\} \leq \min\{u_i^{(d)}, u_j^{(d)}\} + \varepsilon_c, \qquad d = 1, \ldots, n.$$

Only exact overlap or contact directly yields a contained box corridor. Tolerance-gap, segment, and repair edges are search aids; they support a certificate only after conversion to a conservatively validated box corridor or finite-cover OBB corridor.

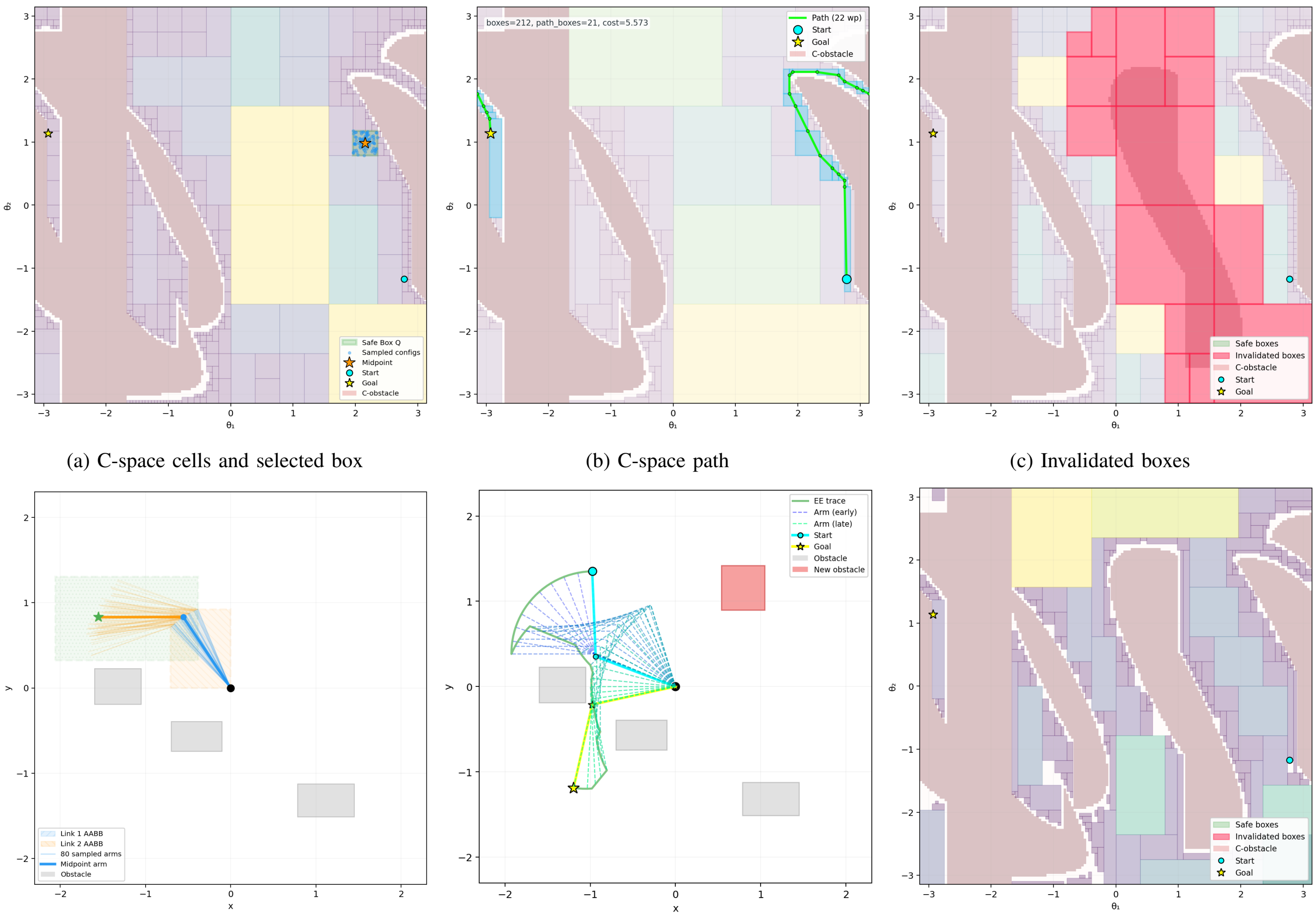
B. Build Pipeline

Let $G_{\cap} = (\mathcal{F}, E_{\cap})$ denote the reusable subgraph of exact overlaps and contacts. Algorithm 1 implements this canonical-address build: it batch-validates a LECT-aligned frontier, compresses $\mathrm{CertFree}$ cells by exact-union coalescing, and refines unresolved regions through an anchor-prioritized budget. The operation $\widehat{V}_{\mathcal{O}}(I; \kappa_E, \delta)$ obtains matched or materialized endpoint evidence, derives current link envelopes, applies $V_{\mathcal{O}}$, and maps missing or inconclusive evidence to $\mathrm{Unknown}$. The reusable output comprises $\mathcal{F}$, $G_{\cap}$, and deferred frontier state; query attachments and typed corridor transitions are constructed afterward.

C. Corridor Query and Certificate

The typed graph separates candidate origin from route evidence: $G = (\mathcal{F}, E_{\cap} \cup E_{\varepsilon} \cup E_s \cup E_{\mathrm{obb}} \cup E_{\mathrm{rep}})$, comprising overlap/contact, tolerance gap, segment, finite-cover OBB-corridor, and repair edges. Edge types record how candidate corridors were produced; certificate status is determined solely by Definition 1 and the witness construction below.

For a returned graph route, an overlap/contact transition contributes its incident validated boxes and contained gateway segment; a validated OBB transition contributes its retained finite cover. A tolerance-gap, segment, repair, or endpoint attachment contributes no witness until converted to one of these region covers. Concatenating their parameter windows



(a) C-space cells and selected box

(b) C-space path

(c) Invalidated boxes

(d) Workspace AABBs

(e) Workspace path

(f) After regrowth

Fig. 2. Two-link RBF build–query–update schematic. Top: (a) selected C-space box, (b) route, and (c) invalidation; bottom: (d) workspace link AABBs, (e) mapped route, and (f) regrowth. Embedded counts and costs are illustrative; θ_0, θ_1 are joint coordinates and x, y are workspace coordinates; end effector (EE) and waypoint (wp) are abbreviated in panels (b) and (e).

constructs W; the route is certificate-backed exactly when the windows cover $[0, 1]$ without an unsupported transition.

A best-first traversal uses a heuristic priority. For overlap routes, gateways lie in consecutive box intersections, and convexity keeps their connecting segments within witnessed boxes.

Lemma 1 (Box-corridor certificate)**.** *For scene revision ν_{scene} and descriptor δ, let $\mathcal{F}_{\text{cert}}$ contain boxes labeled CertFree against $\mathcal{O}(\nu_{\text{scene}})$, each directly validated or retained as a zero-gap exact union of such boxes under the same pair. Set $R_{\text{cert}} = \bigcup_{B \in \mathcal{F}_{\text{cert}}} B$. If a continuous path $\gamma : [0, 1] \to \mathcal{Q}$ lies in R_{cert}, then $\gamma([0, 1]) \subseteq \mathcal{C}_{\text{free}}(\mathcal{O}(\nu_{\text{scene}}))$ under Assumption 1.*

Proof. Directly validated boxes lie in $\mathcal{C}_{\text{free}}$ by Corollary 1; each coalesced box is an exact union of such boxes under the same $(\nu_{\text{scene}}, \delta)$. Hence $R_{\text{cert}} \subseteq \mathcal{C}_{\text{free}}$, and the path inclusion follows. □

Corollary 3 (Route construction from an overlap corridor)**.** *Let $B_1, \ldots, B_m \in \mathcal{F}_{\text{cert}}$, with $q_s \in B_1$, $q_g \in B_m$, and gateways $g_i \in B_i \cap B_{i+1}$. The polyline $(q_s, g_1, \ldots, g_{m-1}, q_g)$ lies in R_{cert} and is collision-free under Assumption 1. Replacing a route segment by a continuous segment whose complete image is covered by its retained, validated finite OBB cover preserves the conclusion. Any transition not converted to a retained validated box or finite-cover OBB witness makes the returned route validation-required.*

Proof. Let $g_0 = q_s$ and $g_m = q_g$. For each $i = 1, \ldots, m$, both endpoints of $[g_{i-1}, g_i]$ lie in the convex box B_i, so the complete polyline lies in R_{cert} and Lemma 1 applies. A validated OBB replacement follows from Corollary 2. □

VI. Experiments

The theory above establishes the soundness contract; the evaluation addresses four questions about operating cost and planning behavior. Q1 characterizes a fixed warm-cache Shelf+KUKA LBR iiwa (Shelf+IIWA) profile and compound profile controls; Q2 provides path-quality context against roadmap, sampling, and convex-region baselines; Q3 characterizes selected in-sample profiles on three robots; and Q4 isolates endpoint- and link-envelope trade-offs. Fig. 3 introduces the fixed Shelf scene. The Q2 baseline couples iterative regional inflation with Graphs of Convex Sets (IRIS/GCS).

Algorithm 1 Zero-gap exact coalescing and budgeted refinement.

Input: Anchors $\mathcal{S}$, LECT $\mathcal{T}_L$, endpoint identity κ_E, obstacles $\mathcal{O}$, current revision ν_{scene}, policy $V_{\mathcal{O}}$, descriptor δ, depths $0 \leq D_0 \leq D_s$, refinement-admission cap $B_{\max} \in \mathbb{Z}_{\geq 1}$, and time, depth, and retention caps
Output: Labeled boxes $\mathcal{F}$, overlap subgraph $G_\cap$, deferred frontier domains $\mathcal{C}$

form from $\mathcal{S}$ and $\mathcal{T}_L$ the identity-deduplicated frontier through depths $D_0{:}D_s$, then batch-evaluate it: $s_I \leftarrow \widehat{V}_{\mathcal{O}}(I;\kappa_E,\delta)$
for each candidate family $\mathcal{G}$ of CertFree dyadic cells with matching revision and descriptor, set $H \leftarrow \operatorname{bbox}(\bigcup_{B\in\mathcal{G}} B)$
coalesce only if dyadic integer ranges establish $\operatorname{ExactUnion}(H,\mathcal{G})$
insert H with CertFree and verified-tiling status under $(\nu_{\text{scene}},\delta)$; consume constituents only from the active node set
insert unmerged $(I,\text{CertFree})$ leaves into $\mathcal{F}$; build $G_\cap$
initialize $\mathcal{P}$ with maximal remaining frontier cells, keyed by region identity and prioritized by anchor coverage; $\mathcal{C} \leftarrow \emptyset$; $b \leftarrow 0$
while $\mathcal{P}$ is nonempty, $b < B_{\max}$, and time/depth caps remain **do**
  admit and batch-evaluate at most $B_{\max} - b$ cells; increment b by the batch size
  for each evaluated I: commit it and update $G_\cap$ if $s_I = \text{CertFree}$; otherwise enqueue unseen LECT-aligned children if I is splittable, or defer I into $\mathcal{C}$
end while
assign the retention-capped union of $\mathcal{C}$ and $\mathcal{P}$ to $\mathcal{C}$; record omissions
return $\mathcal{F}, G_\cap, \mathcal{C}$

Table III maps each empirical question to its evidence and reuse mechanism.

Protocol and metrics. Experiments ran on Ubuntu 22.04 with an Intel Core i5-12600KF; OMPL planners were single-threaded, and imported timings without host-load metadata are descriptive.

Build denotes post-initialization reusable construction. For RBF, it includes evidence access or materialization, scene labeling, graph/anchor construction, and shortcuts, but excludes snapshot creation/opening; PRM endpoints are preloaded. Online/q denotes per-query work. Timings exclude simplification and the independent audit, which densifies each polyline to at most 0.01 rad. Time coordinates are method-local, whereas path ratios use query-matched references. Because the artifacts omit complete W, audit-passing returns are not reported as certificate-backed.

Q1 Batch task is one completion within a jointly scheduled five-query run (40 values from eight runs). Q3 uses

$$\text{Task/q} = \left(T^{\text{wall}}_{\text{adapt}} + \sum_{i=1}^{N_q} T^{\text{internal}}_{\text{query},i} \right) / N_q.$$

Here N_q is batch size; the two terms are scene-adaptation wall time and internal query time. Reuse curves add Build/K to a fixed task coordinate; Shelf $K > 5$ and random-scene $K > 10$ are extrapolated. Tables report success-only medians and interquartile ranges. L is joint-space path length, and $L_{\text{ref},\lambda}$ is the pooled empirical lower envelope of audit-passing returns from the fixed methods and checkpoints before selection, not a known optimum or held-out reference; Shelf uses its per-query counterpart. For Fig. 4, let $\bar{L}$ and $\bar{L}_{\text{ref}}$ denote the displayed audited mean and its minimum reference.

Profile inheritance and cache accounting. Q1 and Q2 use a read-only depth-23 LECT and a 100-box admission cap (d23/b100). Q3 inherits the RBF build/query backend, collision checker, postprocessor, and 0.01-rad audit, but not the Shelf anchors, b100 cap, or Shelf cache root. Its six robot–difficulty profiles use robot-local roots: IIWA and Panda replay depth-23 and depth-20 records, respectively, whereas UR5 materializes endpoint evidence. Warm Build includes lookup, reads, and demand paging but excludes snapshot creation/opening, storage, peak memory, and the one-time d23 prewarm (16.8 million records, 12.74 GiB, 1008.6 s). Q1 compares endpoint-source/replay and link-representation profiles; Q4 separately measures endpoint construction and isolated link-envelope operations.

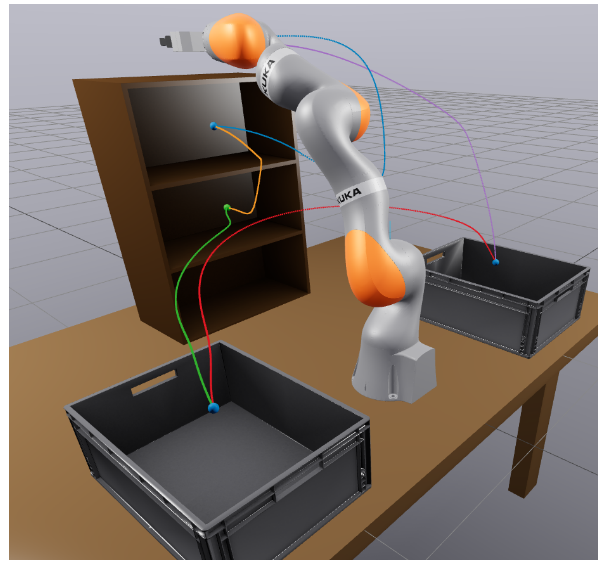


Fig. 3. Canonical Marcucci shelf scene [5] with a KUKA LBR iiwa manipulator (Shelf+IIWA) and the five fixed anchors.

A. Q1: Shelf+IIWA Replay and Profile Controls

Fig. 3 shows the fixed cycle through approach-side (AS), target-shelf (TS), center-shelf (CS), left-bin (LB), and right-bin (RB), executed as AS–TS–CS–LB–RB–AS. Table IV reports the b100 profile over seeds 0–7; b100 was selected during development on this cycle and then held fixed for Q1 and Q2. *Takeaway.* All 40 warm-d23 IFK-AA/SupportHull baseline paths pass the 0.01-rad audit, with median Build and within-batch task-completion diagnostics of 0.171 and 0.077 s. The HIFK-5/no-cache row changes both endpoint construction and replay; its 2.770-s Build (16.2×) is a compound profile contrast, not a cache-only effect. Critical-sample replays its matching d23 cache, whereas Link-AABB and SupportHull-only retain IFK-AA+d23. Table IV reports the controls.

B. Q2: Shelf Cross-Algorithm Context

RBF, IRIS/GCS, and PRM use reusable structures configured from the fixed five-query set; RRT-Connect and BIT* run independently per query. PRM shares one endpoint-preloaded cumulative roadmap per seed. Operating points use all five queries and eight runs per method.

For Table V, PRM and BIT* independently select, for each query, the lowest-median-time checkpoint among checkpoints with audit-passing returns for all eight seeds and an audited mean path within 1% of that query's best. IRIS/GCS+post instead selects the fastest 40/40 prefix within 8% of its best aggregate ratio. Fig. 4 uses separate method-level 8% display markers; the RBF marker is the registered b100 point. All selectors use the reported Shelf runs, so Q2 compares selector-conditioned path quality, not held-out performance or cross-planner latency. A common postprocessor determines path statistics; its time is excluded.

Takeaway. At the selected operating points, RBF has lower median query-normalized ratios than PRM and RRT-Connect for all five queries, but higher ratios than BIT* for all five and IRIS/GCS+post for four.

C. Q3: Random Scenes Across Robot Models

Dataset and selection. Q3 uses a frozen catalog of three robots, ten seeds per difficulty, and ten queries per scene: 60 Medium/Hard scenes and 600 saved scene–query instances per method. For each robot–seed pair, Hard strictly extends Medium by an obstacle-prefix addition; queries are shared and direct start–goal segments are obstructed. The obstacle prefixes and one RBF profile per robot–difficulty condition were selected on this same catalog. Eligibility required 100/100 audit-passing returns within that condition; among eligible candidates, selection used Task/q and path length. Q3 therefore characterizes six selected in-sample operating points.

Compared methods. RBF reuses one query-agnostic forest per ten-query scene. PRM builds one endpoint-preloaded cumulative roadmap per scene; RRT-Connect and BIT* are query-local. Full-success RRT-Connect and PRM curves advance after $\geq 1\%$ aggregate path-ratio gains. Table VI selects BIT*'s fastest per-instance checkpoint within 8% of its best path, whereas Fig. 5 uses the 1% display rule.

Results. Median path ratios are 0.02–0.31 lower than RRT-Connect in all six conditions and lower than PRM in five; UR5 Medium is 0.03 higher. They are also lower than BIT* in five; Panda Medium is 0.01 higher. Holding Task/q fixed, the recorded $\mathrm{Build}/K + \mathrm{Task/q}$ accounting projection decreases by 69.6–79.7% from $K = 1$ to $K = 10$ across the three robot-wise means.

D. Q4: Endpoint and Link-Envelope Mechanisms

Tables VII and VIII report Q4 over six joint-interval widths. Each study uses a deterministic set of 1000 IIWA joint boxes per width, shared across all representations within that study. The endpoint study reports median construction time and volume. Gap is the negative of the largest per-axis width shortfall to the per-box extent hull of the Critical-sample, Analytical, and Monte Carlo references; more negative is worse.

The link study reports mean isolated envelope-build and separation costs rather than production-cascade latency. SupportHull's AABB broadphase and all early exits are disabled. Core volume is the zero-radius centerline-envelope volume B_ℓ° before radius lifting.

Takeaway. At 0.50 rad, HIFK-5 has normalized endpoint volume 2.07 versus 8.61 for IFK-AA (76% lower) at 9.8 times the endpoint-construction cost. Across the tested widths, isolated AABB overlap takes 4.7–6.1% of the mean per-probe time required by direct SupportHull/GJK.

E. Supplementary Scene-Cache Update Timing

Table IX records supplementary scene-cache update timings outside Q1–Q4; planning is excluded and brackets are interquartile ranges.

F. Discussion and Scope

Three components form one algorithmic interface: canonical dyadic addresses couple endpoint replay with forest refinement; exact-union coalescing compresses the current graph without enlarging the verified set; and typed box/OBB transitions compose corridor witnesses. Cache depth and graph size remain decoupled because the admission budget exposes only anchor-relevant cells while primitive keys remain unchanged by revision-local unions. Changing endpoint construction changes κ_E, and changing link representation changes δ, but neither changes route-status semantics.

RBF targets repeated queries for a fixed robot under local scene revisions. Q2 reports path quality, Q3 reports selected in-sample profiles, and the controls quantify warm-replay and mechanism costs. Complete route-witness serialization remains future work.

VII. Conclusion

RBF turns obstacle-independent kinematic bounds into reusable planning state. A shared dyadic identity maps each endpoint record to the primitive cell of a rebuilt scene graph, while compositional corridor verification combines current-scene witnesses and exposes typed repair interfaces. The budgeted builder refines canonical cells and coalesces only exact dyadic unions, preserving the validated free set while reducing the revision-local graph. On Shelf+IIWA, 40/40 warm-profile paths passed the sampled audit and median warm Build was 0.171 s. On the six-condition in-sample profile-selection catalog, selected profiles had lower median query-matched path ratios than RRT-Connect in all six and than PRM and BIT* in five. With Task/q held fixed, the recorded $\mathrm{Build}/K + \mathrm{Task/q}$ accounting projection decreases by 69.6–79.7% from $K = 1$ to $K = 10$. The soundness analysis establishes a cross-revision reuse contract, while the experiments quantify operating costs and selected path-quality behavior. Isolating end-to-end speedup across a scene revision remains future work.

TABLE III
EVALUATION QUESTIONS, EVIDENCE, AND REUSE MECHANISMS

Question	Tested claim	Main evidence	Reuse mechanism
Q1 Shelf profile	Characterize fixed warm-cache cost and compound profile controls.	Eight-seed controls on a five-query cycle.	Endpoint replay and scene-local rebuild.
Q2 Shelf comparison	Contextualize audited path quality and within-method trade-offs.	Five methods over eight seeds.	Within-scene graph reuse over five queries.
Q3 Random scenes	Describe selected-profile path quality across robot models.	Six robot–difficulty conditions, 100 queries each.	Within-scene graph reuse over ten queries.
Q4 Mechanisms	Isolate endpoint- and link-envelope trade-offs.	1000 shared boxes per width at six widths.	Endpoint construction and link-envelope validation.
Cache update	Record scene-cache maintenance primitives.	Historical insertion, removal, and warm rebuild.	Supplementary timing outside Q1–Q4.

TABLE IV
SHELF+IIWA RBF D23 BASELINE AND PROFILE CONTROLS

Case	Build	Batch-task diag.	Diag. 5q	$L_q/L_{\mathrm{ref},q}$
RBF baseline	0.171 [0.166, 0.174]	0.077 [0.072, 0.085]	0.111 [0.107, 0.117]	1.16 [1.09, 1.40]
Critical-sample	0.279 [0.275, 0.283]	0.087 [0.078, 0.095]	0.145 [0.134, 0.151]	1.16 [1.10, 1.38]
Link-AABB	0.167 [0.164, 0.171]	0.083 [0.077, 0.086]	0.116 [0.111, 0.119]	1.21 [1.09, 1.50]
HIFK-5/no cache	2.770 [2.644, 3.003]	0.181 [0.139, 0.207]	0.739 [0.703, 0.777]	1.16 [1.10, 1.44]
SupportHull-only	0.193 [0.188, 0.200]	0.087 [0.082, 0.095]	0.126 [0.120, 0.136]	1.16 [1.09, 1.40]

Notes: Seconds; medians [25%, 75%]; b100, 8 seeds. Batch task: 40 within-batch timestamps, not single-query latency; ratios: matching audits. Baseline: IFK-AA+d23+Link-AABB+SupportHull/GJK. Diag. 5q $=$ Build/5 + Batch task.

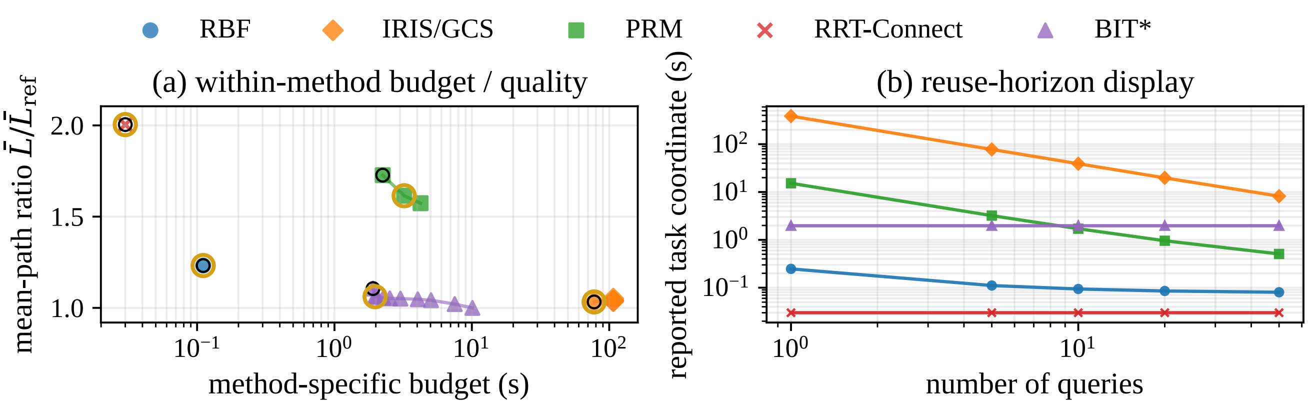


Fig. 4. Shelf+IIWA trade-offs over eight seeds. Left: audited mean-path ratio $\bar{L}/\bar{L}_{\mathrm{ref}}$. Black rings mark the first 40/40 budget; gold rings mark the registered RBF profile and each comparison method's fastest full-success point within 8% of its best mean. Right: selected method-local task coordinates across K ($K > 5$ extrapolated). Table V gives the Q2 selections; postprocessing/audit is excluded.

TABLE V
SHELF+IIWA CROSS-ALGORITHM COMPARISON OF AUDITED PATHS

	RBF b100 Build 0.171 s		**IRIS/GCS+post** Build 385.950 s		**PRM** Build 20.040 s		**RRT-Connect** No reusable build		**BIT*** No reusable build	
Query	Batch task (s)	$L_q/L_{\mathrm{ref},q}$	T (s)	$L_q/L_{\mathrm{ref},q}$	T (s)	$L_q/L_{\mathrm{ref},q}$	T (s)	$L_q/L_{\mathrm{ref},q}$	T (s)	$L_q/L_{\mathrm{ref},q}$
AS→TS	0.065 [0.064, 0.070]	1.40 [1.34, 1.65]	0.540 [0.390, 0.661]	1.15 [1.11, 1.18]	0.198 [0.147, 0.248]	1.61 [1.42, 1.73]	0.034 [0.024, 0.052]	2.27 [2.05, 2.72]	4.004 [4.002, 4.004]	1.06 [1.02, 1.08]
TS→CS	0.085 [0.082, 0.089]	1.13 [1.11, 1.14]	0.346 [0.296, 0.396]	1.22 [1.11, 1.28]	0.173 [0.118, 0.312]	2.99 [2.11, 3.22]	0.054 [0.021, 0.089]	3.98 [2.80, 5.09]	10.138 [10.004, 10.386]	1.09 [1.04, 1.11]
CS→LB	0.073 [0.070, 0.077]	1.50 [1.35, 1.85]	0.411 [0.314, 0.518]	1.03 [1.02, 1.05]	0.136 [0.120, 0.195]	1.53 [1.42, 2.13]	0.032 [0.027, 0.070]	2.24 [1.79, 2.56]	7.504 [7.504, 7.725]	1.05 [1.04, 1.08]
LB→RB	0.078 [0.075, 0.083]	1.10 [1.08, 1.24]	0.761 [0.685, 0.879]	1.02 [1.01, 1.06]	0.164 [0.091, 0.399]	1.20 [1.11, 1.29]	0.001 [0.001, 0.002]	1.31 [1.29, 1.34]	1.654 [1.654, 1.654]	1.03 [1.03, 1.04]
RB→AS	0.081 [0.077, 0.086]	1.05 [1.04, 1.05]	0.537 [0.418, 0.609]	1.00 [1.00, 1.00]	0.252 [0.096, 0.295]	1.24 [1.18, 1.25]	0.001 [0.001, 0.001]	1.08 [1.04, 1.10]	0.204 [0.204, 0.204]	1.02 [1.01, 1.03]

Notes: Seconds; medians [25%, 75%], eight seeds. RBF Batch task: 40 within-batch completions from eight five-query runs; other T values are per query. Times are method-local; postprocessing/audit is excluded.

TABLE VI
PROFILE-SELECTION-CATALOG SUMMARIES AT THE REPORTED OPERATING POINTS

Scenario	RBF			PRM			RRT-Connect		BIT*	
	Build	Task/q	$L/L_{\mathrm{ref},\lambda}$	Build	Online/q	$L/L_{\mathrm{ref},\lambda}$	Online/q	$L/L_{\mathrm{ref},\lambda}$	Selected Online/q	$L/L_{\mathrm{ref},\lambda}$
IIWA-Medium	0.075 [0.058, 0.083]	0.009 [0.007, 0.011]	1.00 [1.00, 1.03]	3.567 [3.007, 3.933]	0.038 [0.014, 0.070]	1.03 [1.02, 1.12]	0.002 [0.001, 0.002]	1.02 [1.01, 1.05]	0.014 [0.007, 0.042]	1.05 [1.03, 1.07]
IIWA-Hard	0.097 [0.086, 0.105]	0.014 [0.011, 0.015]	1.00 [1.00, 1.03]	4.175 [3.408, 5.637]	0.048 [0.021, 0.102]	1.03 [1.02, 1.08]	0.002 [0.002, 0.003]	1.03 [1.00, 1.07]	0.027 [0.012, 0.076]	1.05 [1.03, 1.07]
UR5-Medium	0.262 [0.254, 0.270]	0.075 [0.020, 0.086]	1.05 [1.00, 1.14]	3.124 [2.966, 3.418]	0.323 [0.085, 0.426]	1.02 [1.00, 1.12]	0.023 [0.002, 0.049]	1.23 [1.06, 1.54]	0.026 [0.012, 0.058]	1.06 [1.03, 1.14]
UR5-Hard	0.276 [0.264, 0.284]	0.083 [0.043, 0.108]	1.05 [1.01, 1.14]	2.488 [2.325, 3.048]	0.419 [0.129, 0.437]	1.06 [1.01, 1.19]	0.048 [0.008, 0.090]	1.36 [1.14, 1.58]	0.046 [0.022, 0.091]	1.07 [1.03, 1.17]
Panda-Medium	0.250 [0.218, 0.265]	0.028 [0.013, 0.039]	1.05 [1.02, 1.09]	3.431 [3.364, 4.180]	0.066 [0.038, 0.159]	1.06 [1.03, 1.08]	0.005 [0.002, 0.011]	1.22 [1.12, 1.35]	0.016 [0.007, 0.031]	1.04 [1.02, 1.06]
Panda-Hard	0.224 [0.221, 0.237]	0.043 [0.021, 0.053]	1.05 [1.01, 1.16]	2.337 [1.627, 3.978]	0.054 [0.034, 0.126]	1.10 [1.05, 1.16]	0.056 [0.019, 0.136]	1.25 [1.14, 1.37]	0.062 [0.027, 0.124]	1.06 [1.04, 1.12]

Notes: Seconds; medians [25%, 75%]. RBF Task/q: ten-query batch coordinate; OMPL $\mathrm{Online}/\mathrm{q}$: 100 queries/condition. BIT* is selected per instance; UR5 Medium uses query-local edges; postprocessing/audit is excluded.

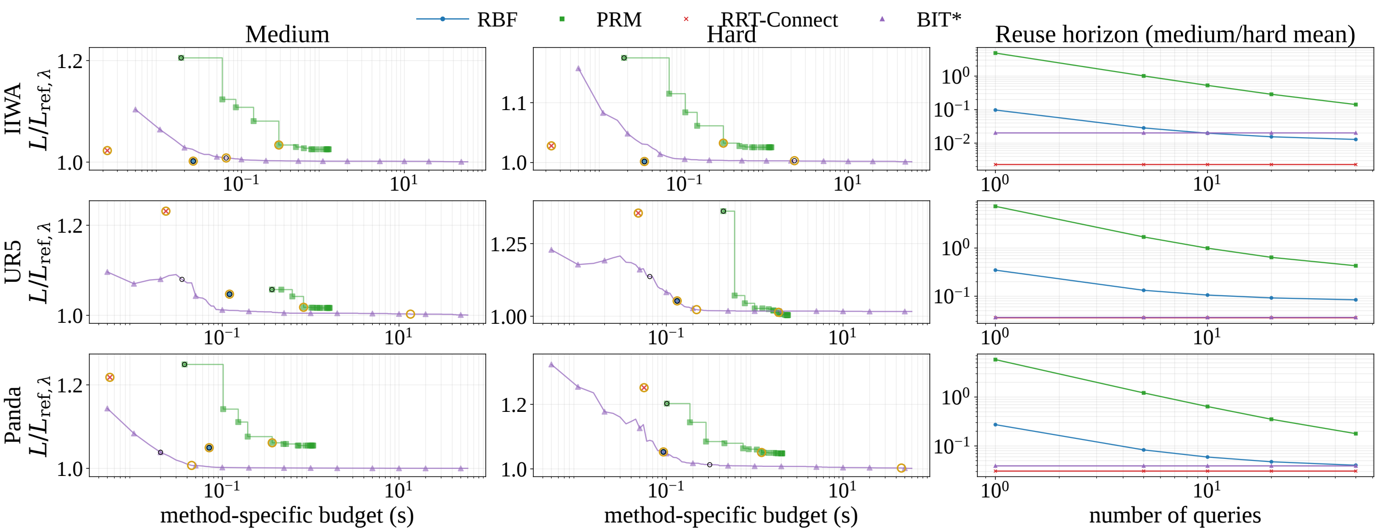


Fig. 5. Random-scene trade-offs. Left: OMPL traces and reported RBF profiles at $K = 5$. Black rings mark first full-catalog success; gold rings mark reported RBF and comparison-method display selections, which advance after $\geq 1\%$ path-ratio gains. Right: selected task coordinates averaged over Medium/Hard across K ($K > 10$ extrapolated). Table VI gives query-matched ratios; postprocessing/audit is excluded.

TABLE VII
ENDPOINT-ENVELOPE EVALUATION OVER JOINT-INTERVAL SIDE LENGTHS (RADIANS)

Source	Volume ratio						Time (μs)						Gap (mm)					
	0.02	0.05	0.10	0.20	0.30	0.50	0.02	0.05	0.10	0.20	0.30	0.50	0.02	0.05	0.10	0.20	0.30	0.50
IFK-AA	1.09	1.23	1.52	2.30	3.53	8.61	5.48	6.11	6.45	6.58	6.81	7.35	–	–	–	–	–	–
HIFK-3	1.04	1.10	1.22	1.51	1.89	3.12	25.94	26.85	29.16	27.21	29.27	29.38	–	–	–	–	–	–
HIFK-5	1.02	1.07	1.13	1.30	1.51	2.07	70.40	71.69	73.82	73.15	75.92	72.32	–	–	–	–	–	–
Critical-sample	1.00	1.00	1.00	1.00	0.99	0.98	8.68	9.04	10.03	11.56	12.69	16.33	-0.05	-0.30	-1.17	-4.53	-10.77	-30.97
Analytical	1.00	1.00	1.00	1.00	1.00	1.00	6429.96	6500.41	6602.25	6800.10	6927.12	7096.01	0.00	0.00	0.00	0.00	0.00	-4.49
Monte Carlo	0.64	0.70	0.74	0.78	0.79	0.82	2251.74	5629.24	11233.26	22352.95	33813.58	56835.57	-6.05	-14.62	-32.68	-65.57	-88.87	-123.53

Notes: 1000 boxes/width; medians; volume/Analytical. Gap is the negative largest per-axis width shortfall to the per-box reference extent hull; more negative is worse. Only IFK-AA/HIFK are candidates for the theoretical conservative branch.

TABLE VIII
LINK-ENVELOPE EVALUATION OVER JOINT-INTERVAL SIDE LENGTHS (RADIANS)

Envelope	Core volume (m^3)						Build (μs)						Test (μs)					
	0.02	0.05	0.10	0.20	0.30	0.50	0.02	0.05	0.10	0.20	0.30	0.50	0.02	0.05	0.10	0.20	0.30	0.50
Link AABB	0.0136	0.0201	0.0384	0.147	0.508	4.61	0.138	0.152	0.167	0.166	0.171	0.178	0.586	0.605	0.601	0.604	0.630	0.790
SupportHull	0.000359	0.00266	0.0143	0.106	0.449	4.54	0.205	0.214	0.218	0.219	0.217	0.250	12.520	12.850	12.267	12.223	12.659	13.026

Notes: Means over 1000 boxes/width and five repeats; no early exits. AABB uses overlap; SupportHull direct GJK; Build excludes endpoints.

TABLE IX
SUPPLEMENTARY SCENE-CACHE UPDATE TIMINGS AND WARM-BUILD/UPDATE RATIOS (OUTSIDE Q1–Q4)

Operation	Time (ms)	Warm build/update
Warm build (10 obstacles)	922.75 [897.57, 955.67]	–
Insert $10 \to 15$	6.06 [5.69, 6.54]	153.3–207.5×
Remove $15 \to 10$	0.35 [0.31, 1.65]	943.6–3354.0×
Warm build (15 obstacles)	958.79 [941.67, 1115.18]	–

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